Kalman Filter + Localization

Robots in Context

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Agenda

- Programming Exercise 5 (A*)
  1. Jacob
  2. Alexander
- Programming Exercise 6.1(RTT*)
  - Emil
- PE 6.2 (RRT), volunteers:
  -



Content

- Motivation
- Probabilistic Estimation
- Localization
- Kalman Filter (KF)
  - Simple Observer without covariances
  - Kalman filter
- Next:
  Extended Kalman Filter (EKF) + Particle Filter (PF)



We continue at: 9:15



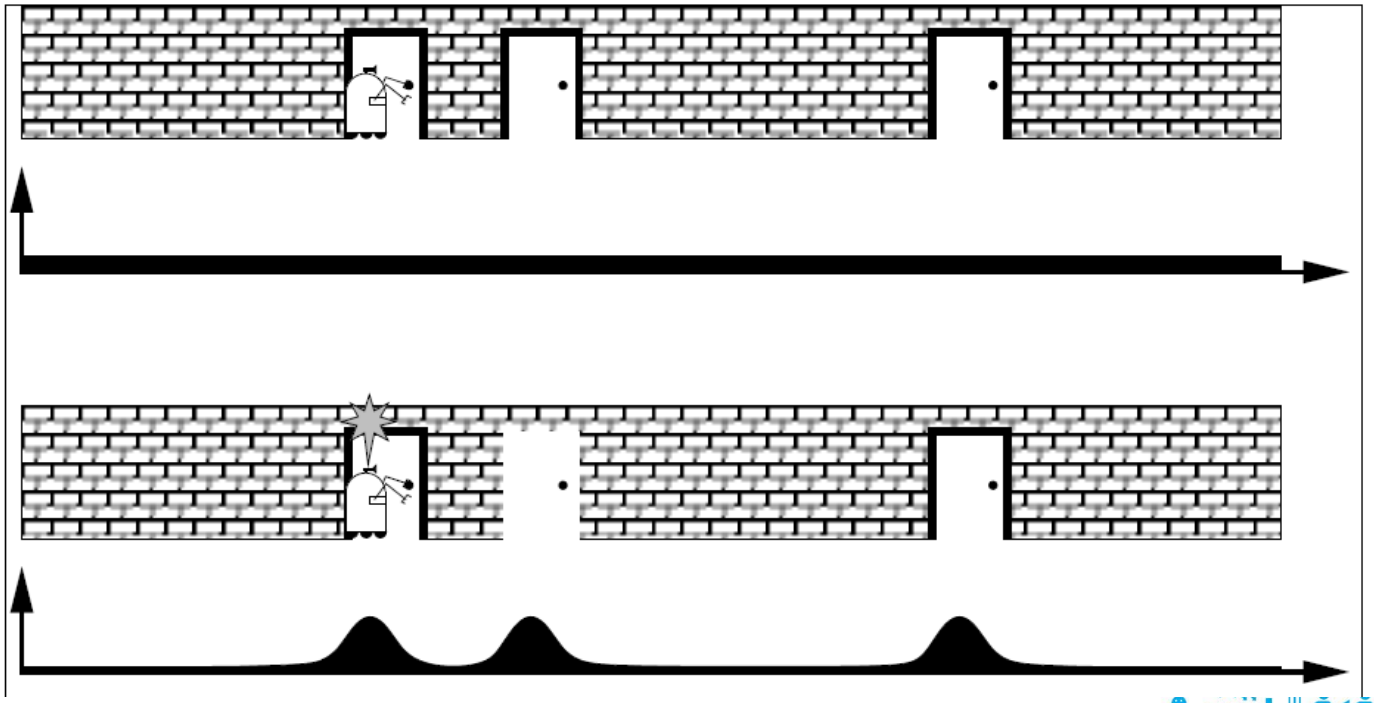
Motivation

- Planners generally assume to have perfect knowledge of the robot state and the environment.
- However, knowledge derived from sensors is imperfect and subject to noise.
- *Probabilistic Estimation* provides a way to deal with imperfect information.
- Kalman filters provide a recursive (and optimal under certain conditions) method for estimating the state based on noisy data.



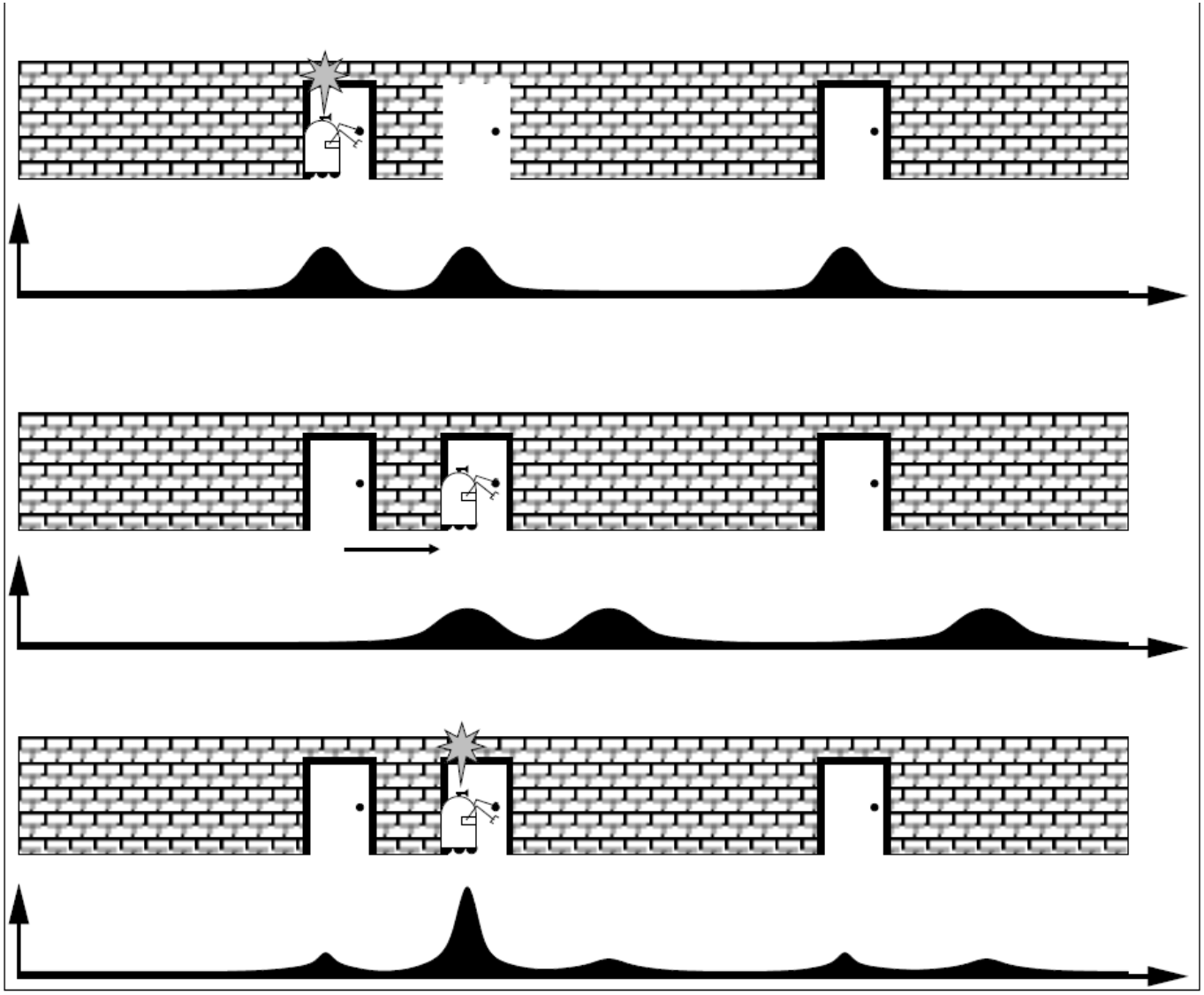
Probabilistic Estimation

- The state is considered as a probability distribution rather than precise values.



[figure: Principles of Robot Motion, Choset et.al.]

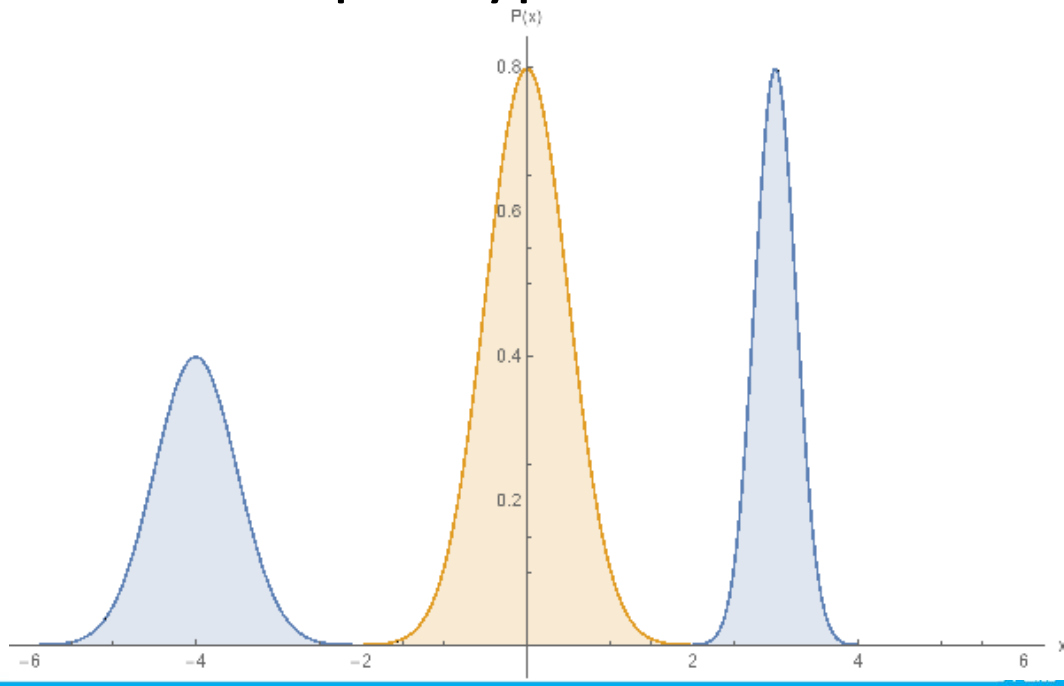
Probabilistic Estimation



[figure: Principles of Robot Motion, Choset et.al.]

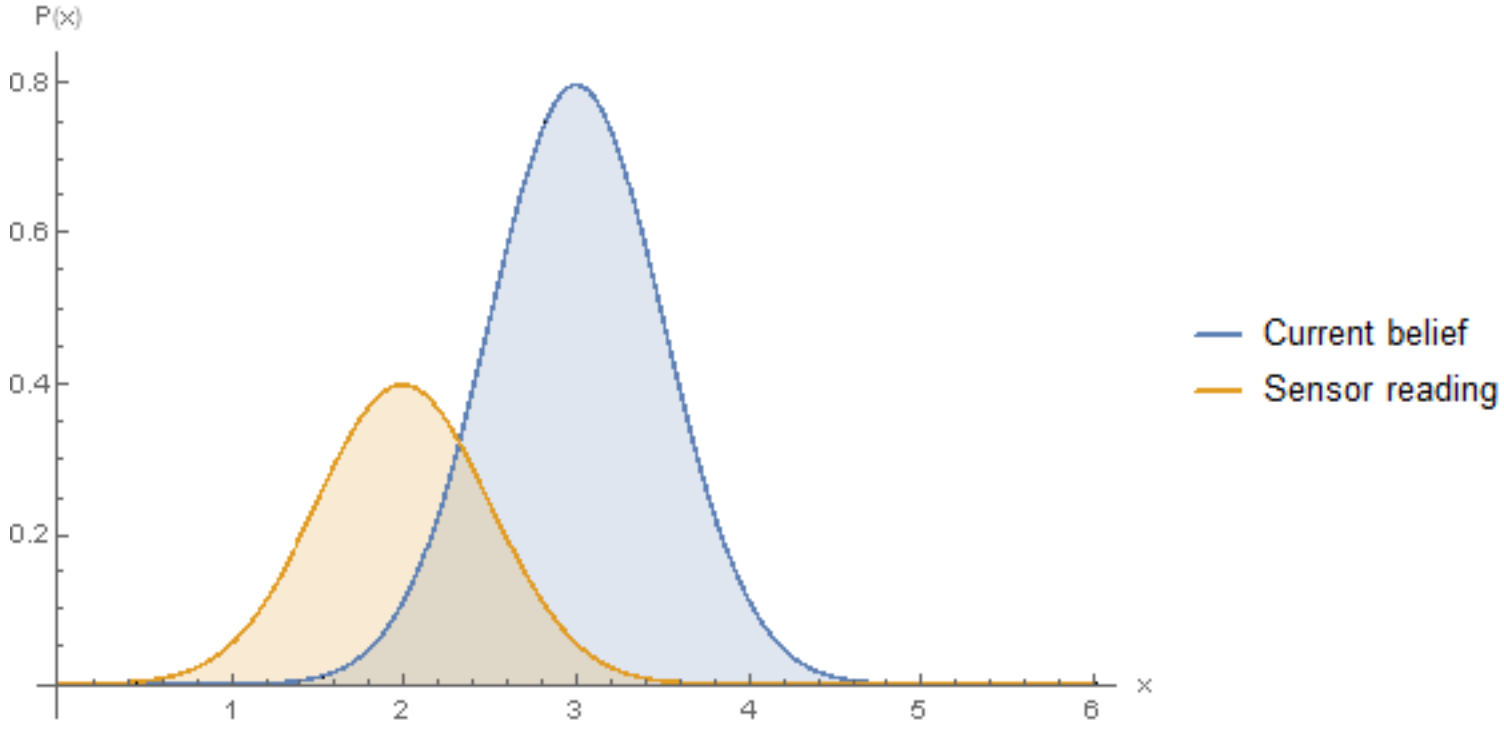
Probabilistic Estimation

- Single vs. Multi Modal
  - Single Modal: One hypothesis
  - Multi Modal: Multiple hypotheses.

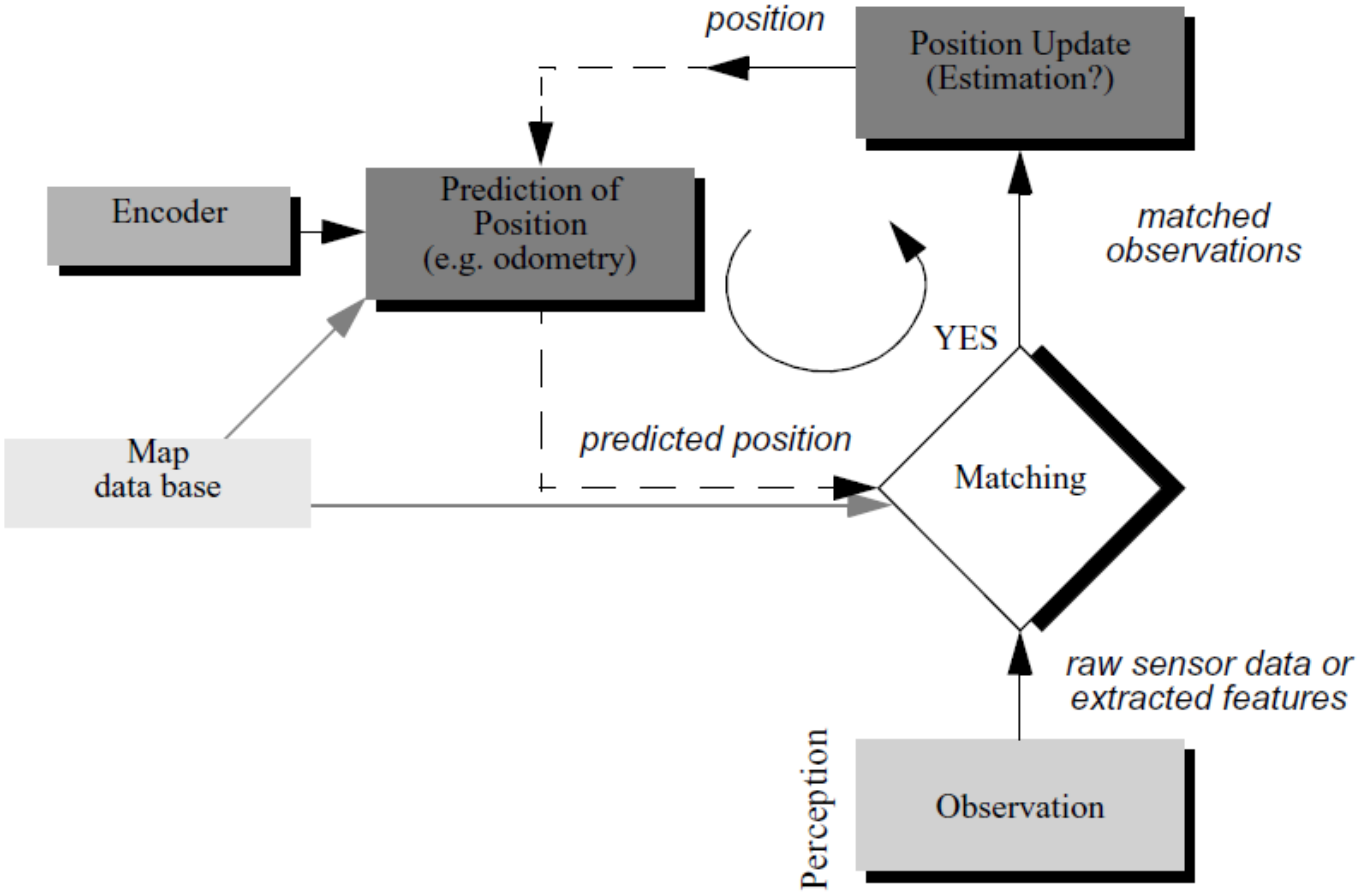


Single Modal Update

- Given a current belief and a sensor reading we wish to find the best believe of where we are.

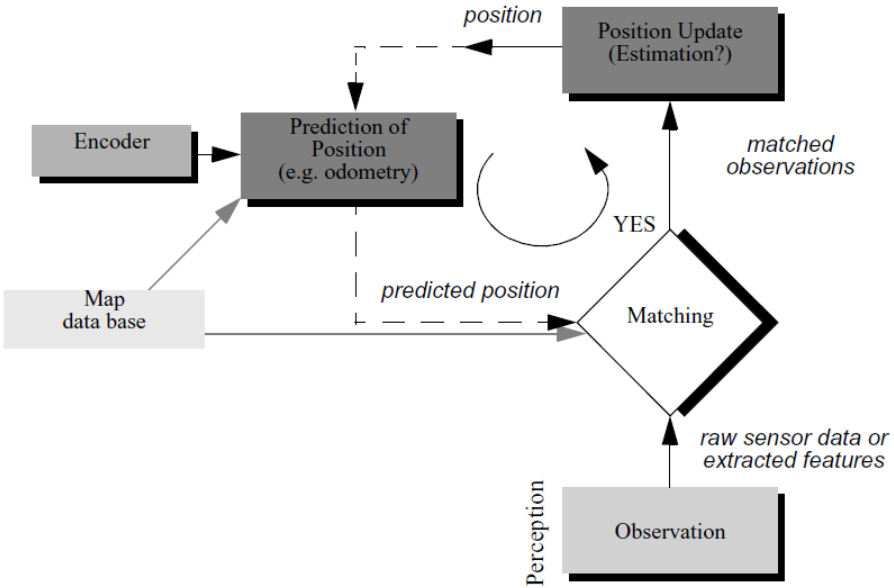


Localization



Kalman Filter

- Assumptions
  - Single Modal Distribution
  - Noise is Gaussian
- Requirements
  - Define the state
  - System model
  - Observation model



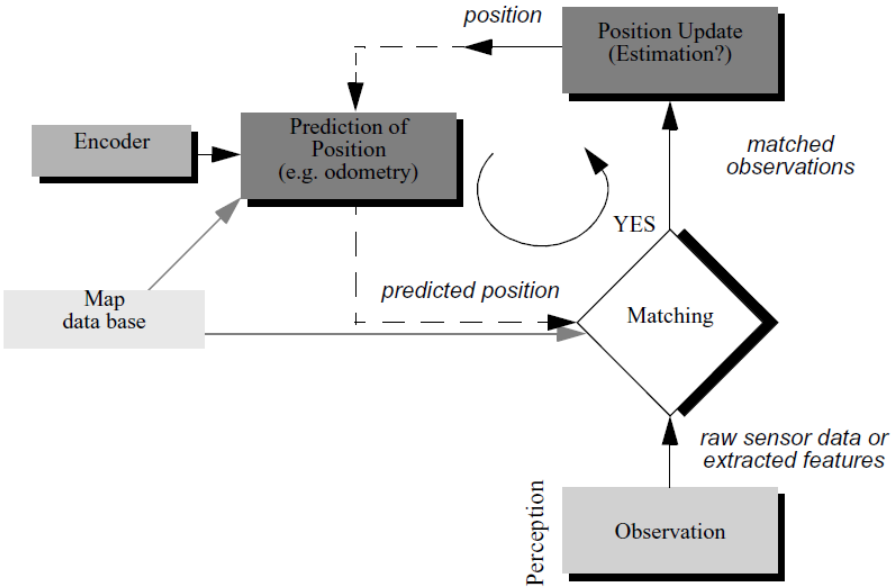
Defining the State

- Observable State
  - E.g. Position and Orientation (x, y, θ)
- Non-observable State
  - E.g. Linear and angular velocity (x', y', θ')
  - Could also include accelerations.
- What to select as the state, $x(k)$, depends on
  - What you need
  - Your System model
  - Your Observation model



Kalman Filter

- Assumptions
  - Single Modal
  - Noise is Gaussian
- Requirements
  - Define the state
  - **System model**
  - Observation model



Linear System Model

$$x(k+1) = F(k)x(k) + G(k)u(k) + v(k)$$

$x(k)$ Vector with state at time k

$F(k)$ Matrix relating current to future state

$u(k)$ Vector with control input at time k

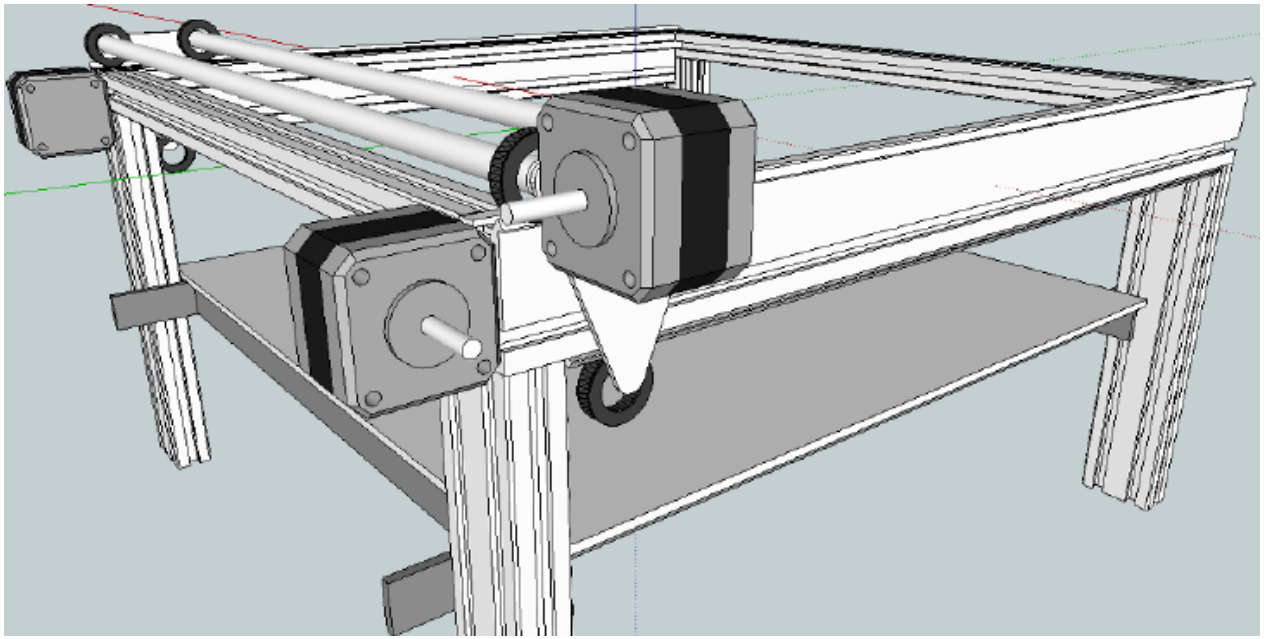
G(k) Matrix relating input to the dynamics

$v(k)$ Vector with process noise with zero mean and covariance V(k)



Example of Linear System Model

XYZ Gantry System



Example of Linear System Model

- System Model $x(k+1) = F(k)x(k) + G(k)u(k) + v(k)$

$$\boldsymbol{x} = \begin{pmatrix} x \\ y \\ z \\ \dot{x} \\ \dot{y} \\ \dot{z} \end{pmatrix} \quad \mathbf{u} = \begin{pmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{pmatrix} \quad \boldsymbol{F}(k) = \begin{bmatrix} 1 & 0 & 0 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t & 0 \\ 0 & 0 & 1 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \boldsymbol{G}(k) = \begin{bmatrix} \frac{1}{2}\Delta t^2 & 0 & 0 \\ 0 & \frac{1}{2}\Delta t^2 & 0 \\ 0 & 0 & \frac{1}{2}\Delta t^2 \\ \Delta t & 0 & 0 \\ 0 & \Delta t & 0 \\ 0 & 0 & \Delta t \end{bmatrix}$$

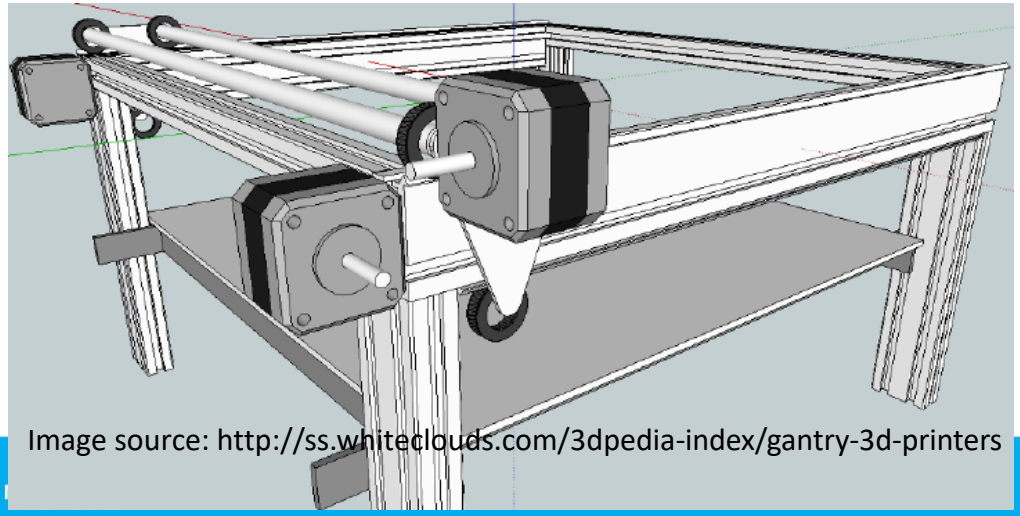
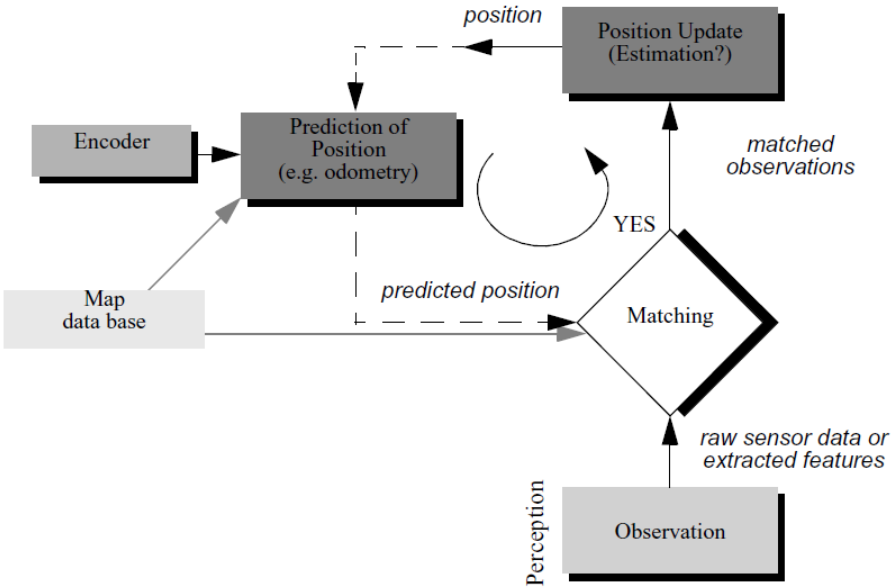


Image source: http://ss.whiteclouds.com/3dpedia-index/gantry-3d-printers

Kalman Filter

- Assumptions
  - Single Modal
  - Noise is Gaussian
- Requirements
  - Define the state
  - System model
  - **Observation model**



Linear Observation Model

$$y(k) = H(k)x(k) + w(k)$$

$y(k)$ Vector representing sensor readings

$x(k)$ Vector with state at time k

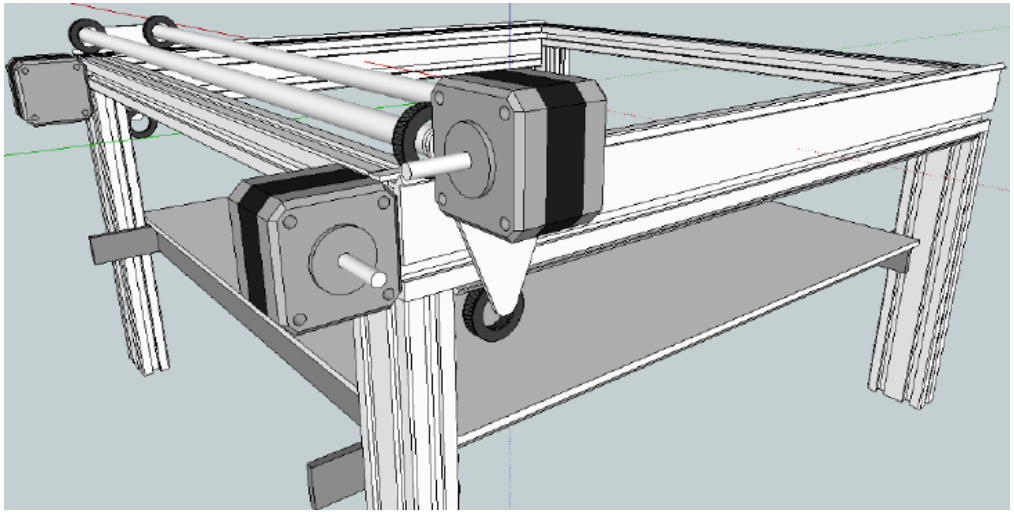
H(k) Matrix mapping state to sensor readings

$w(k)$ Vector with measurements noise with zero mean and covariance W(k)



Example of Linear Observation Model

- Ideas?



Example of Linear Observation Model

- Ideas?
  - Encoders
  - 1D distance sensors
  - UWB
  - GNSS

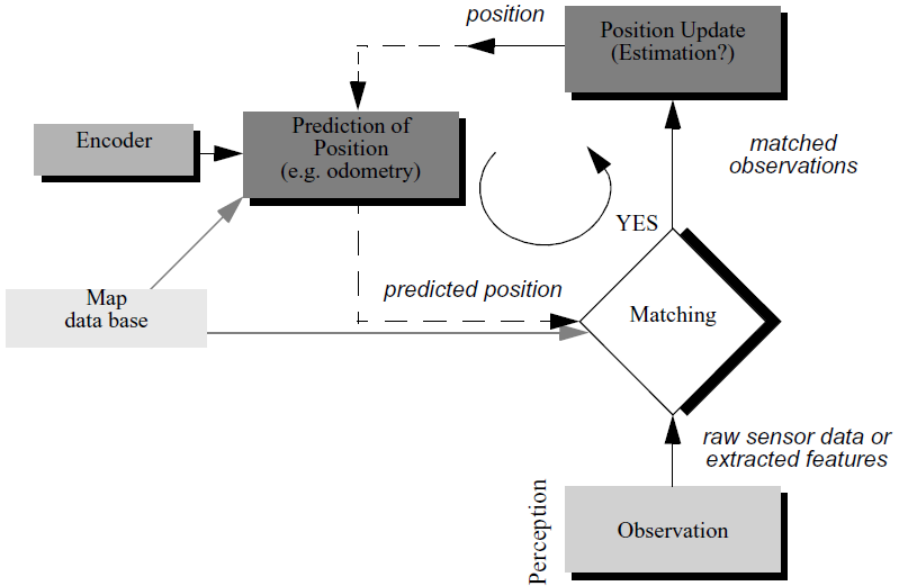
$$\mathbf{y}(k) = \boldsymbol{H}(k)\,\boldsymbol{x}(k)$$

$$\boldsymbol{x} = \begin{pmatrix} x \\ y \\ z \\ \dot{x} \\ \dot{y} \\ \dot{z} \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \boldsymbol{H}(k) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$



(Linear) Kalman Filter

- Insert the System Model and Observation Model
- Add the probability estimation by including covariance matrices.



Linear System Model in Kalman Filter

Consider the linear system without noise

$$x(k+1) = F(k)x(k) + G(k)u(k)$$
$$y(k) = H(k)x(k)$$

Notation

$$\hat{x}(k_1|k_2)$$

This means the state estimate at time k_1 after using observations up to time k_2. Notice that $k_1 \geq k_2$.

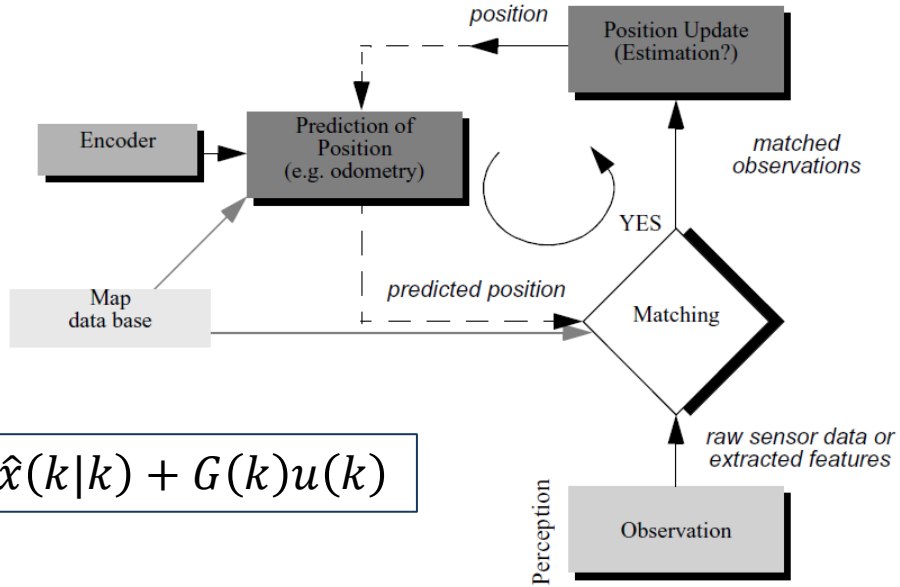
We use $\hat{x}$ to denote the estimate of x.



Linear System Model in Kalman Filter

Given a state $\hat{x}(k|k)$ and control input $u(k)$ the next state can be predicted as:

$$\hat{x}(k+1|k) = F(k)\hat{x}(k|k) + G(k)u(k)$$



$$\hat{x}(k+1|k) = F(k)\hat{x}(k|k) + G(k)u(k)$$

Linear Observer

For observations $y(k+1)$ we have

$$y(k+1) = H(k+1)x(k+1|k)$$

From this we can derive an expression for how to correct the state estimate

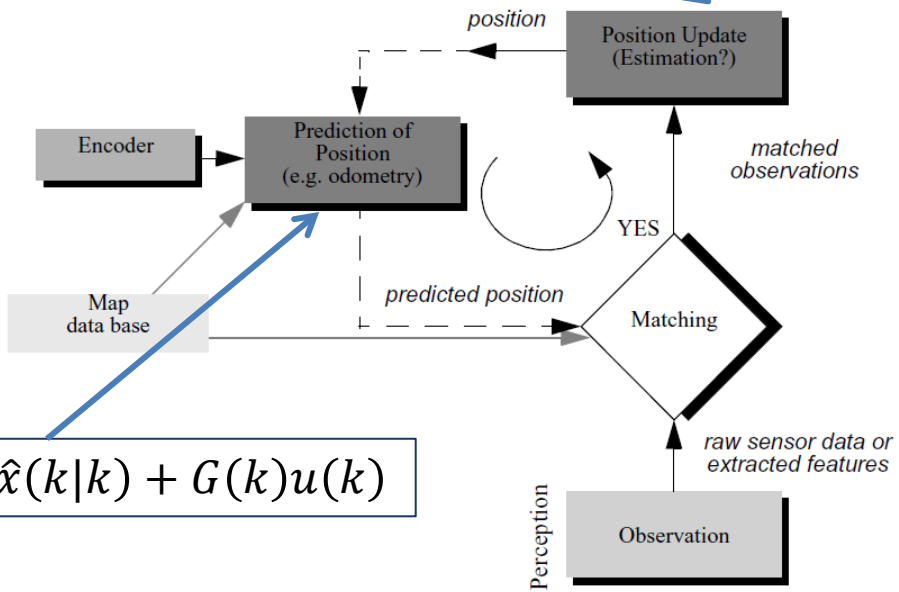
$$\hat{x}(k+1|k+1) = \hat{x}(k+1|k) + H^T(HH^T)^{-1}(y(k+1) - H\hat{x}(k+1|k))$$

Where $H = H(k+1)$



Linear Observer

$$\hat{x}(k+1|k+1) = \hat{x}(k+1|k) + H^T(HH^T)^{-1}(y(k+1) - H\hat{x}(k+1|k))$$



$$\hat{x}(k+1|k) = F(k)\hat{x}(k|k) + G(k)u(k)$$

Using Probability Distributions

- Initially we discussed that a state is not just a state, it is a probability distribution.
- For a state estimate $\hat{x}(k|k)$ we therefore also have a covariance $P(k|k)$
- The Kalman Filter updates both state estimate $\hat{x}$ and covariance P.
  - Assumes noise with zero mean
  - Work ”best” with Gaussian noise



Covariance

- Variance: A measure for how much a variable, X, ”changes”.

$$Var(X) = E[(X - \mu)^2]$$

- Covariance: A measure for how much two or more variables, X, Y, changed together.

$$\mathrm{cov}(X, Y) = E[(X - \mu_x)(Y - \mu_y)]$$

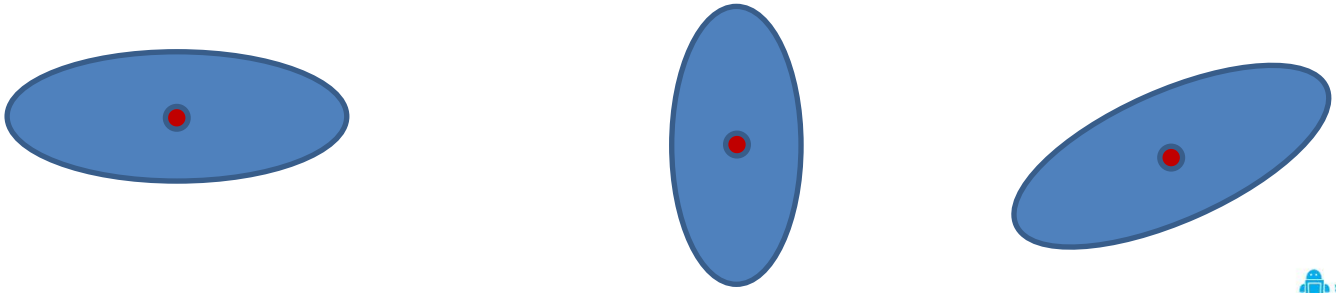


Covariance Matrix

- Given matrix: Matrix with the covariance for all variables in a vector X.

$$\Sigma = \begin{bmatrix} \mathrm{E}[(X_1 - \mu_1)(X_1 - \mu_1)] & \mathrm{E}[(X_1 - \mu_1)(X_2 - \mu_2)] & \cdots & \mathrm{E}[(X_1 - \mu_1)(X_n - \mu_n)] \\ \mathrm{E}[(X_2 - \mu_2)(X_1 - \mu_1)] & \mathrm{E}[(X_2 - \mu_2)(X_2 - \mu_2)] & \cdots & \mathrm{E}[(X_2 - \mu_2)(X_n - \mu_n)] \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{E}[(X_n - \mu_n)(X_1 - \mu_1)] & \mathrm{E}[(X_n - \mu_n)(X_2 - \mu_2)] & \cdots & \mathrm{E}[(X_n - \mu_n)(X_n - \mu_n)] \end{bmatrix}$$

- In 2D/3D often visualized as ellipses or ellipsoids



Kalman Filter - Prediction

- The system model includes noise

$$x(k+1) = F(k)x(k) + G(k)u(k) + v(k)$$

- The noise ”adds” to the covariance

$$P(k+1|k) = F(k)P(k|k)F(k)^T + V(k)$$

Where $V(k)$ is the covariance of $v(k)$.

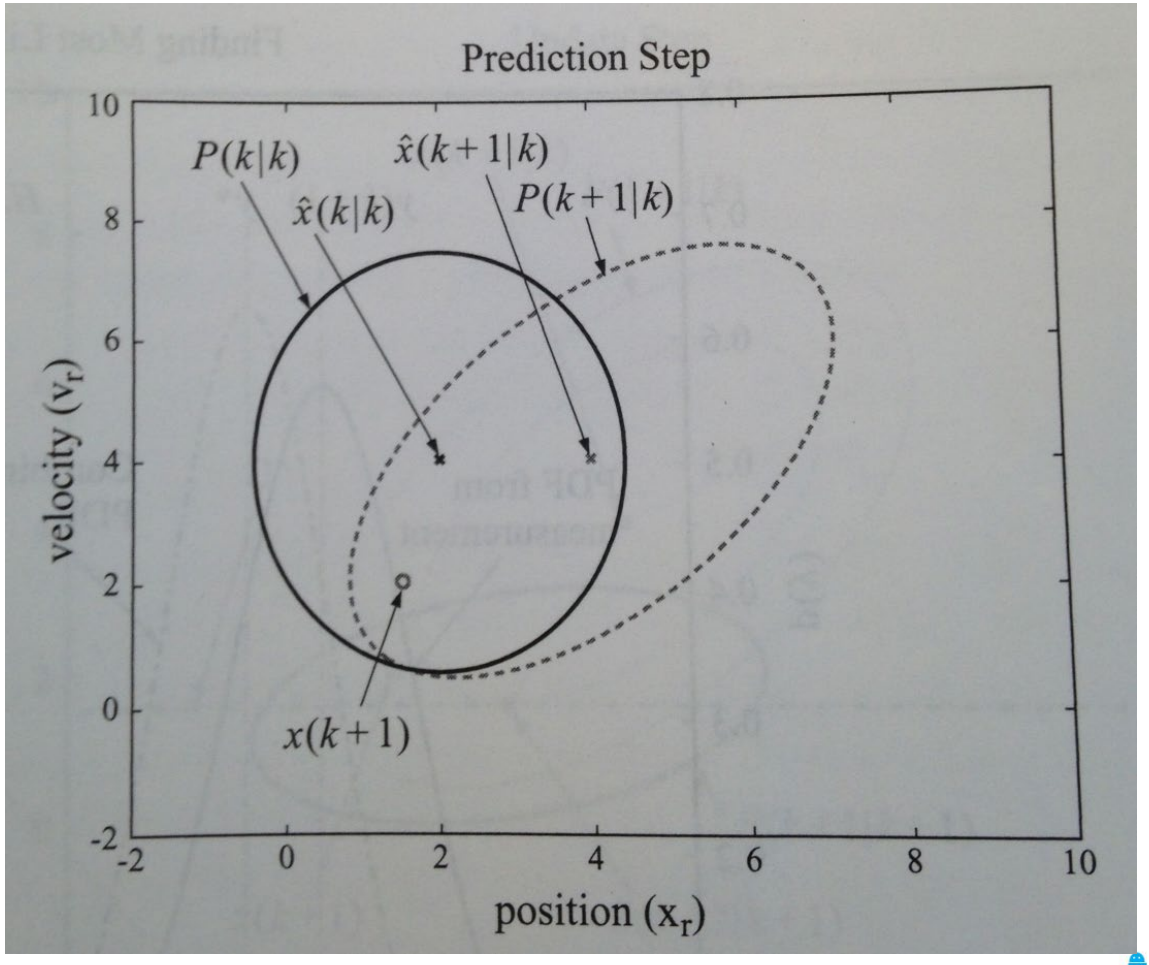
The complete prediction step then becomes

$$\hat{x}(k+1|k) = F(k)\hat{x}(k|k) + G(k)u(k)$$

$$P(k+1|k) = F(k)P(k|k)F(k)^T + V(k)$$

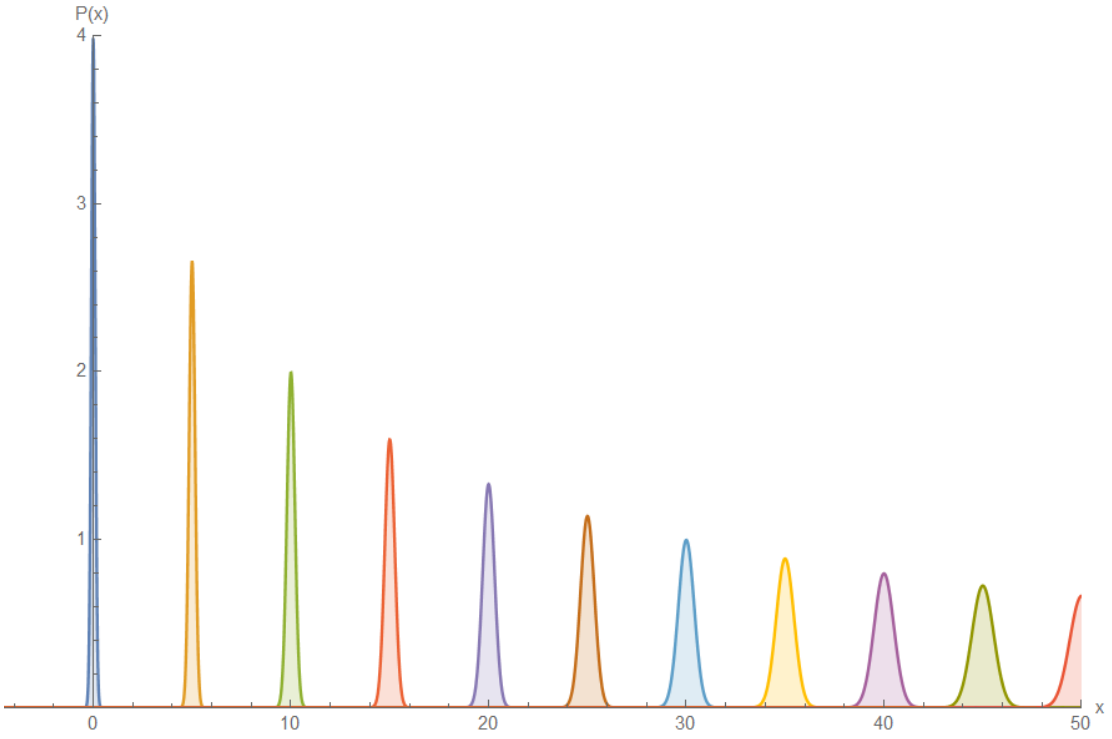


Kalman Filter Prediction



Predicting multiple steps

1D example of how the uncertainty grows



Update with Covariance

- Update of state and covariance

$$\hat{x}(k+1|k+1) = \hat{x}(k+1|k) + R\,v$$
$$P(k+1|k+1) = P(k+1|k) - RHP(k+1|k)$$

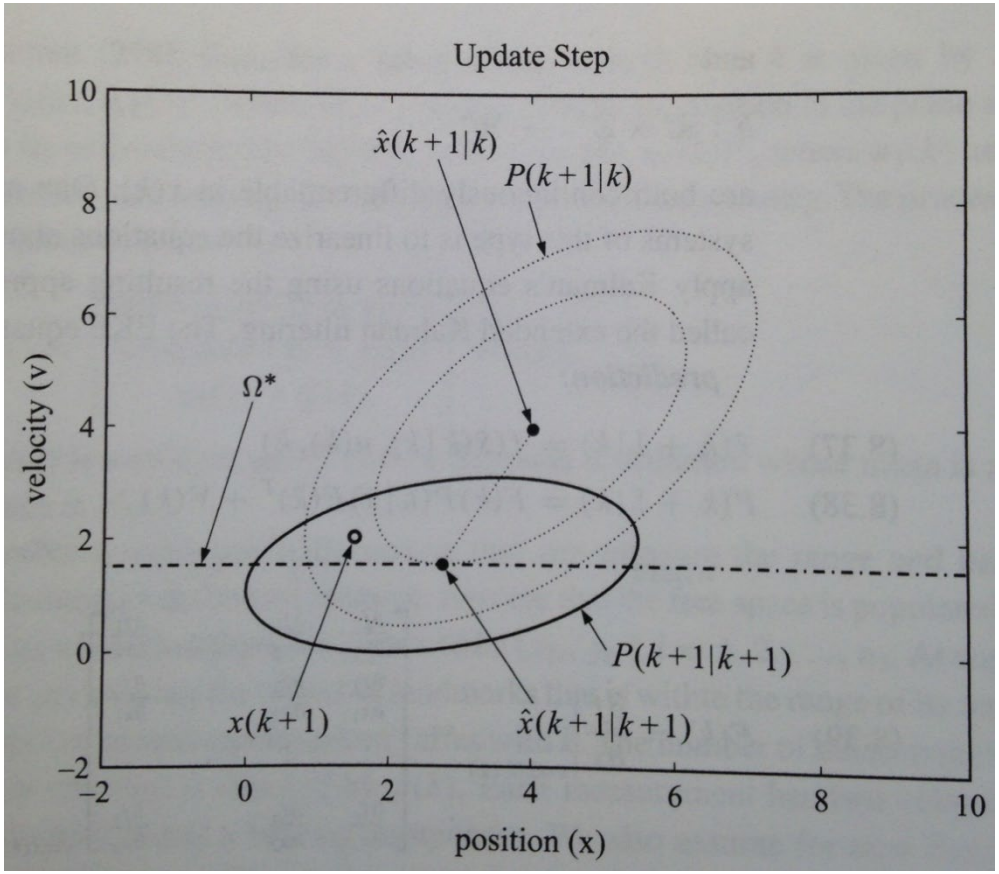
Where

$$v = y(k+1) - H\hat{x}(k+1|k)$$
$$R = P(k+1|k)H^T S^{-1}$$
$$S = HP(k+1|k)H^T + W(k+1)$$

Where $W(k+1)$ is the covariance of the sensor noise.



Update with Covariance



Localization using Kalman Filter

- Assuming control input is given as accelerations and we only wish to know the 2D position, what will then be a reasonable state and control input?

$$x(k) = ?$$
$$u(k) = ?$$



Localization using Kalman Filter

- Assuming control input is given as accelerations and we only wish to know the 2D position, what will then be a reasonable state and control input?

$$x(k) = \{p_x, p_y, \dot{p_x}, \dot{p_y}\}$$
$$u(k) = \{\ddot{p_x}, \ddot{p_y}\}$$



Localization using Kalman Filter

- Given the state and control inputs:

$$x(k) = \{p_x, p_y, \dot{p_x}, \dot{p_y}\}$$

$$u(k) = \{\ddot{p_x}, \ddot{p_y}\}$$

- What should the system model look like?

$$x(k+1) = F(k)x(k) + G(k)u(k) + v(k)$$



Localization using Kalman Filter

- Given the state and control inputs:

$$x(k) = \{p_x, p_y, \dot{p_x}, \dot{p_y}\}$$
$$u(k) = \{\ddot{p_x}, \ddot{p_y}\}$$

- What should the system model look like?

$$x(k+1) = F(k)x(k) + G(k)u(k) + v(k)$$

- $F(k) = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $\mathrm{G}(k) = \begin{bmatrix} \frac{1}{2}\Delta t^2 & 0 \\ 0 & \frac{1}{2}\Delta t^2 \\ \Delta t & 0 \\ 0 & \Delta t \end{bmatrix}$ $v(k) = \begin{bmatrix} v_{px} \\ v_{py} \\ v_{px'} \\ v_{py'} \end{bmatrix}$



Localization using Kalman Filter

- Given the state:

$$x(k) = \{p_x, p_y, \dot{p_x}, \dot{p_y}\}$$

- Assuming we have a GPS system, what should the observation model look like?

$$y(k) = H(k)x(k) + w(k)$$



Localization using Kalman Filter

- Given the state:

$$x(k) = \{p_x, p_y, \dot{p_x}, \dot{p_y}\}$$

- Assuming we have a GPS system, what should the observation model look like?

$$y(k) = H(k)x(k) + w(k)$$

$$H(k) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad w(k) = \begin{bmatrix} w_{px} \\ w_{py} \end{bmatrix}$$



Sandbox Exercise: 1D motion

Initialization:

$x(k|k) = 5$

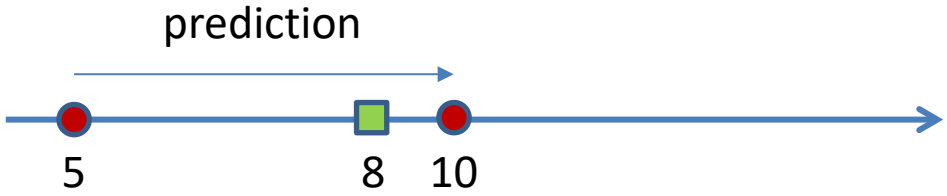
$P = 5$ (state covariance)

$V = 2$ (process noise)

$W = 1$ (measurement noise)

Control input:

$u(k) = 5$



Initial Position

Summary and alternatives

- Kalman Filter
  - Linear model
  - State estimation, handling noise/errors (zero mean)
- Alternatives:
  - Extended Kalman filter: linearizing nonlinear systems
  - Unscented Kalman filter: estimating the covariances



Programming Exercise 8

Consider a simple mobile robot

- Define the system, i.e. the state, system model and observation model (choose something convenient)
- Define a simple scenario, e.g. the robot being directed to drive a predefined pattern but subject to noise.
  - Ensure that the noise is physically meaningful.
- Make sure to keep track of the predicted and the real state. Visualize both.
- Implement a Kalman Filter and inspect the estimated state of the system.
  - Analyze, e.g., how the covariance P evolves (check the trace of the matrix)
- Explore, e.g.,
  - Vary the extent of the noise and/or your estimate of its covariance
  - Add systematic noise, i.e. non-zero mean

